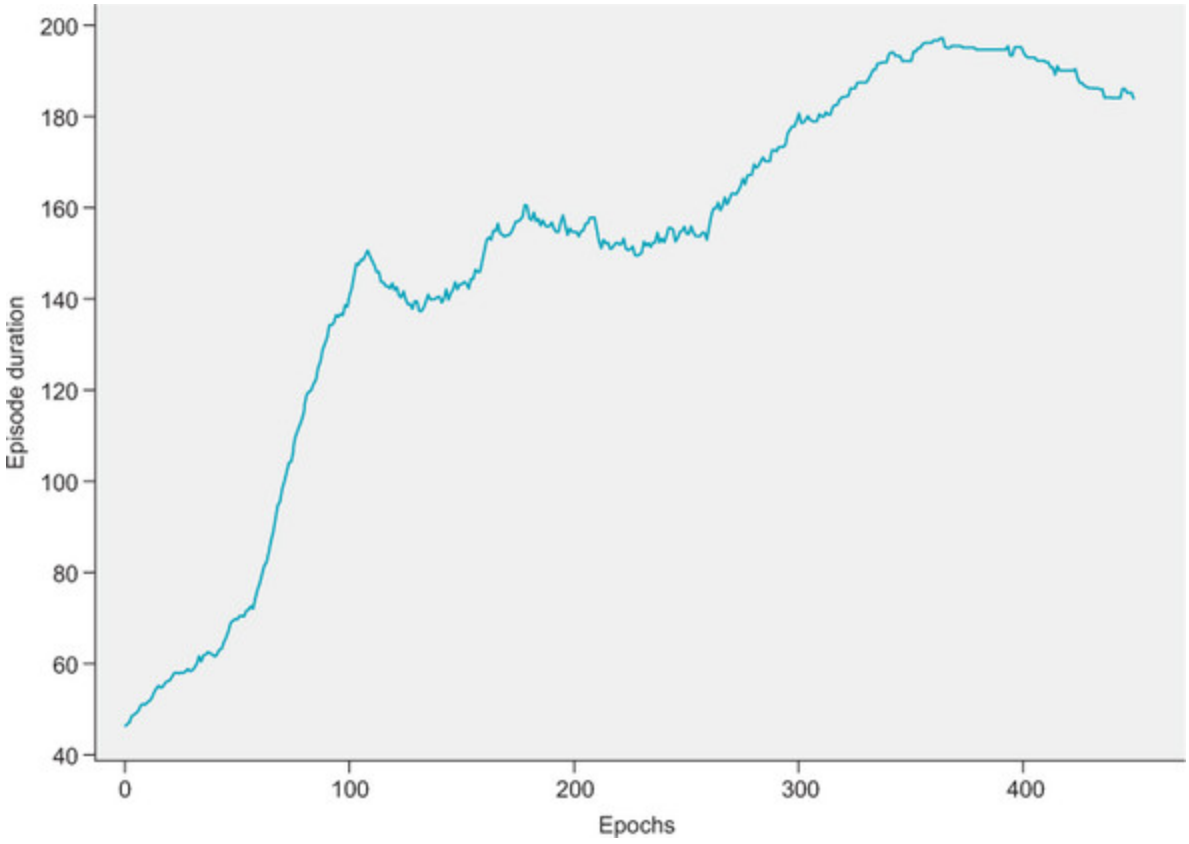
(the state, action, and reward) into separate tensors to train on a batch of data at a time. If you run this code, you should be able to plot the episode duration against the episode number, and you will hopefully see a nicely increasing trend, as in figure 4.10.

Figure 4.10. After training the policy network to 500 epochs, we get a plot that demonstrates the agent really is learning how to play CartPole. Note that this is a moving average plot with a window of 50 to smooth the plot.



The agent learns how to play CartPole! The nice thing about this example is that it should be able to train in under a minute on your laptop with just the CPU. The state of CartPole is just a 4-dimensional vector, and our policy network is only two small layers, so it's much faster to train than the DQN we created to play Gridworld. OpenAI's documentation says that the game is considered "solved" if the agent can play an episode beyond 200 time steps. Although the plot looks like it tops off at around 190, that's because it's a moving average plot. There are many episodes that reach 200 but a few times where it randomly fails early on, bringing the average down a bit.